

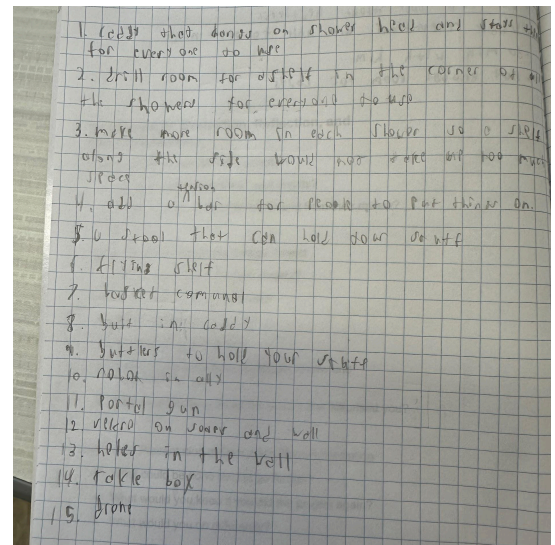
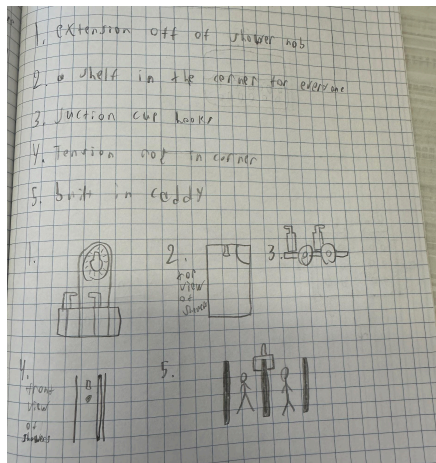
Shower project

Team OH-J

Element D: Design Concept Generation, Analysis, and Selection

In the boys dorms the dorm residents need a place to put their toiletries because there is not a built-in shelf in the communal showers and setting things on the floor is unsanitary. Solving the problem can help the dorm students place their toiletries somewhere without having to set it down on the ground. The problem is that there is nowhere to hang your shampoo and that leads to people putting it on the ground.

For brainstorming we all just came up with ideas that we think would work and then we chose the ones that would actually work later.



1. It will hang off of the shower knob and have a basket to hold all of your toiletries
2. a shelf will be installed in the corner of the showers and it will hold your toiletries
3. A shelf that attaches to the wall with suction cups and can hold your toiletries

Then we used the decision matrix to find out which idea is the best so we added the criteria to each one. The decision matrix let us conclude that the best idea was the built in caddy. We knew from the start that this was a great idea because you keep all the good qualities about the current caddies but now it is improved in every way.

Design Requirements	Current Solution: Caddy that sits on wall	Idea #1: Shelf that hangs from tap	Idea #2: Built in shelf	Idea #3: Suction Cup shelf	Idea #4: Tension Rod	Idea #5: Built in Caddy
Sturdy > 10 lbs	0	0	+1	0	0	+1
Should be able to hold 5 bottles and 1 razor and caddy	0	+1	+1	+1	+1	+1
Must be small. 6 inches by 12 inches can exceed	0	+1	0	0	-1	+1
Lightweight < 2lbs not necessary	0	-1	0	0	-1	0
Easy to clean	0	0	0	0	0	0
Elevated off the floor	0	0	0	0	-1	0
Cost efficient < 10\$ can exceed	0	0	0	0	0	-1
If you bump into it. It should stay up	0	0	0	0	0	+1
Portable. Able to be moved with 1 hand not necessary	0	-1	-1	-1	-1	-1
Total Score		0	1	0	-2	2

Our final solution is the built in shower caddy, it is essentially adding a caddy to each shower so that there is a compartment for everyone to hold their toiletries. We will build caddies that can be placed over top of the shower walls just like the caddies that are used now. The only difference is people will not have to bring their own caddies and there will be community caddies. This is an improvement because people will not have to worry about buying a caddy or if it breaks then they have no other option.

We will make the caddy with a 3D printer that is in the engineering room. We will use TinkerCAD to design it and use the filament that is in the engineering room. The caddy will be

12 x 6 inches on both sides of the caddy. It will be built from the 3D printer and will cost \$20 for one roll of filament.

